

# TAK Server Management

# TAK Server – Logging

- There are many log files that are available for monitoring during execution. They are located within the /opt/tak/logs directory:
  - To view a specific log, you can run the following:
    - **tail -f /opt/tak/logs/<log file>**
      - Example: **tail -f /opt/tak/logs/takserver-messaging.log**
  - Types of log files:
    - **takserver-messaging.log** – Execution-level information about the messaging process, including client connection events, error messages and warnings.
    - **takserver-api.log** – Execution-level information about the API (Application Programming Interface) process, including error messages and warnings.
    - **takserver-messaging-console.log** – Java Virtual Machine (JVM) informational messages and errors, for the messaging process.
    - **takserver-api-console.log** – Java Virtual Machine (JVM) informational messages and errors, for the API process.

# TAK Server – Database Manipulation

- Purging the database will reset the entire database to a clean slate.
  - To purge the TAK Server database run the following:
    - **sudo java -jar /opt/tak/db-utils/SchemaManager.jar purge && sudo java -jar /opt/tak/db-utils/SchemaManager.jar upgrade**
- To start the database:
  - **sudo /opt/tak/db-utils/SchemaManager.jar upgrade**
  - You can run **sudo /opt/tak/db-utils/SchemaManager.jar** to see the different commands it can do.
- To manually archive the database:
  - **sudo /opt/tak/db-utils/archive-data.sh**
    - Note: wherever you run this is where the archive file will be stored.
- To manually restore a database:
  - **sudo /opt/tak/db-utils/restore-data.sh <file>**

# TAK Server – Mission Packages and Files Expiration

- From the Mission Manager you are able to manipulate the files as needed.
- To set an expiration date for a file go to the mission manager (Data → Mission manager):
  - For mission packages and other files, you can set an expiration date by selecting the item then selecting “Edit”.
    - Inside the editing panel, under the “Expiration” tab, you are able to set the date and time for the file/data package to expire at.
  - If you wish to delete the entry immediately you can select the item, then select “Delete” option instead of editing.

# TAK Server – Data Retention Tool

- Data Retention Policy:
  - The Data Retention Tool (Administrative → Data Retention) allows the user to specify the retention policy for the following data types:
    - CoT (no-chat)
    - GeoChat (chat CoT messages)
    - Missions (including mission structure, tracks and files)
    - Mission Packages
    - Files (all enterprise sync data that is not mission data)
  - Users have the option to defines a TTL (time-to-live) interval on data, where the data will expire after it exceeds the TTL.
  - The deletion of the data is handled by the Retention Service that executes periodically.
    - It can be set to a scheduled frequency but is set to never by default.

# Updating the TAK Server

- Update the TAK Server:
  - Download the latest version of TAK Server to your local machine.
    - You can get the latest version from the following link: <https://tak.gov/products/tak-server>
  - Transfer the file to your TAK Server machine:
    - Navigate to where the TAK Server install is on your local machine.
    - Open a terminal at that location.
    - Transfer the file:
      - **scp <TAKServer File> <username>@<IP address>:~/**
        - **Example:** scp takserver-5.2-RELEASE16.noarch.rpm <username>@<IP>:~/
  - On your TAK Server
    - Update the server
      - **sudo dnf install takserver-5.2-RELEASE16.noarch.rpm --setopt=clean\_requirements\_on\_remove=false -y**

```
[takadmin@takserver-el8 ~]$ sudo dnf install takserver-5.0-RELEASE31.noarch.rpm --setopt=clean_requirements_on_remove=false -y
Updating Subscription Management repositories.
Last metadata expiration check: 0:04:39 ago on Sat 13 Jan 2024 05:29:41 AM GMT.
Dependencies resolved.
=====
Package                        Architecture  Version      Repository      Size
=====
Upgrading:
takserver                      noarch       5.0-RELEASE31 @commandline    658 M
=====
Transaction Summary
=====
Upgrade 1 Package

Total size: 658 M
Downloading Packages:
Running transaction check
Transaction check succeeded.
Running transaction test
```

# Updating the TAK Server

- Reload the system manager:
  - **sudo systemctl daemon-reload**
- Restart the TAK Server:
  - **sudo systemctl restart takserver**
- Monitor the logs for errors:
  - **sudo tail -f /opt/tak/logs/takserver-messaging.log**

```
[takadmin@takserver-el8 ~]$ sudo tail -f /opt/tak/logs/takserver-messaging.log
2024-01-13-05:21:03.144 [Thread-2] INFO c.b.marti.nio.netty.NioNettyBuilder - Successfully Started Netty Server for TLS
serverInitializer on Port 8089 with watermark WriteBufferWaterMark(low: 8192, high: 16384)
2024-01-13-05:21:05.336 [main] INFO t.server.federation.FederationServer - Federation disabled in config. Not starting V
2 federation server.
2024-01-13-05:21:06.051 [main] INFO com.bbn.marti.nio.server.NioServer - max memory (bytes): 1736441856
2024-01-13-05:21:06.051 [main] INFO com.bbn.marti.nio.server.NioServer - Server started
2024-01-13-05:21:06.112 [main] INFO t.server.cache.DatafeedCacheHelper - Done initializing PluginDatafeedCacheHelper wit
h buffer cache value: 300 seconds
2024-01-13-05:21:06.125 [main] INFO com.bbn.marti.util.VersionBean - TAK Server version 5.0-RELEASE-29-HEAD
2024-01-13-05:21:06.125 [main] INFO c.b.m.sync.EnterpriseSyncCacheHelper - multicore CPU detected. File cache enabled.
2024-01-13-05:21:06.125 [main] INFO c.b.m.sync.EnterpriseSyncCacheHelper - Initialized EnterpriseSyncCacheHelper with fil
e cache TTL 120 seconds. Implementation: caffeine
2024-01-13-05:21:06.126 [main] INFO tak.server.ServerConfiguration - Started ServerConfiguration in 36.29 seconds (proce
ss running for 73.713)
2024-01-13-05:21:06.134 [main] INFO tak.server.ServerConfiguration - Started TAK Server messaging Microservice.
2024-01-13-05:35:59.263 [main] INFO tak.server.ServerConfiguration - Starting ServerConfiguration v5.0-RELEASE-31 using
Java 17.0.9 with PID 4442 (/opt/tak/takserver.war started by tak in /opt/tak)
2024-01-13-05:35:59.264 [main] INFO tak.server.ServerConfiguration - The following 2 profiles are active: "plugins", "me
ssaging"
2024-01-13-05:36:07.208 [main] INFO tak.server.util.DataSourceUtils - 4 CPU cores detected. Postgres server maximum allo
wed connections value is 2100. The computed connection pool size is: 200
2024-01-13-05:36:10.244 [main] INFO SQL dialect - HHH000400: Using dialect: org.hibernate.dialect.PostgreSQLDialect
2024-01-13-05:36:14.013 [services-deployment-worker-#75%ignite-takserver] WARN com.bbn.marti.service.SSLConfig - TLS en
abled, but no certificate revocation lists, and OSCP is not enabled in Core Config!
2024-01-13-05:36:23.914 [main] INFO c.b.m.sync.JDBCEnterpriseSyncService - multicore CPU detected. File cache enabled.
2024-01-13-05:36:23.918 [main] INFO c.b.marti.service.SubmissionService - input class type: Input
2024-01-13-05:36:23.919 [main] INFO c.b.marti.service.SubmissionService - Configuring network input com.bbn.marti.config
.Input stdssl:
2024-01-13-05:36:24.071 [main] WARN com.bbn.marti.service.SSLConfig - TLS enabled, but no certificate revocation lists,
```